

IS-A Relationship

Inheritance allows a class to derive attributes and methods from another class. It facilitates the "IS-A" relationship (e.g., a Car IS-A Vehicle).

Superclass (Parent): The class being inherited from.

Subclass (Child): The class that inherits from the parent.

1. The 'extends' Keyword

To inherit from a class, use the extends keyword in the class declaration.

```
class Vehicle {  
    protected String brand = "Ford";  
    public void honk() { System.out.println("Beep!"); }  
}  
  
class Car extends Vehicle {  
    private String model = "Mustang";  
}
```

2. Accessing Protected Members

The protected access modifier is key in inheritance. Attributes marked protected are accessible within the **subclass**, even if the subclass is in a different package.

3. Stopping Inheritance

If you wish to prevent other classes from inheriting your class, use the `final` keyword.

```
final class Vehicle { ... }  
  
// This will cause a compilation error:  
class Car extends Vehicle { ... }
```

Why Use Inheritance?

- **Reusability:** Use code from existing classes without rewriting it.
- **Polymorphism:** Allows child classes to share a common parent type.
- **Organization:** Creates a clear hierarchy for complex systems.